



DNA Barcoding and Habitat Preferences of Hermit Crabs (Crustacea: Decapoda) from Chattogram-Cox's Bazar Coast

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ABSTRACT

A total of 170 gastropod shells containing hermit crabs were collected from four sites along the Chattogram-Cox's Bazar coast. Morphological analysis identified three new hermit crab records for Bangladesh, with further DNA barcoding and phylogenetic analysis ongoing. A positive correlation was observed between shell size and hermit crab

abundance, with *I. lacera* being the most commonly used shell. The taxonomic and genetic data from this study will aid marine biologists, ecologists, and conservationists in biodiversity management and conservation efforts.

INTRODUCTION

Hermit crabs, belonging to the infraorder Anomura (Decapoda), are diverse marine organisms, with some species inhabiting estuaries or freshwater. They rely on empty gastropod shells for protection, selecting them based on availability. This study integrates morphological and molecular approaches to explore their biodiversity, phylogenetics, and ecological adaptations in the region.

MOTIVE OF THE RESEARCH

1. To identify hermit crabs and host shells based on their morphological characteristics
2. To classify hermit crabs phylogenetically using DNA barcoding
3. To investigate the habitat preferences of hermit crabs regarding environmental variables

METHODOLOGY

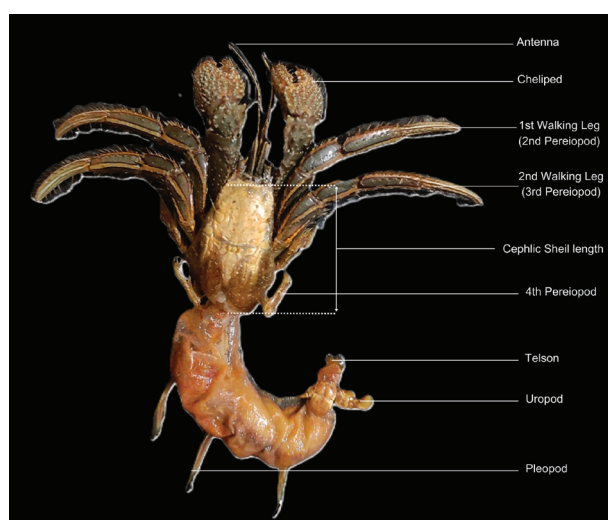
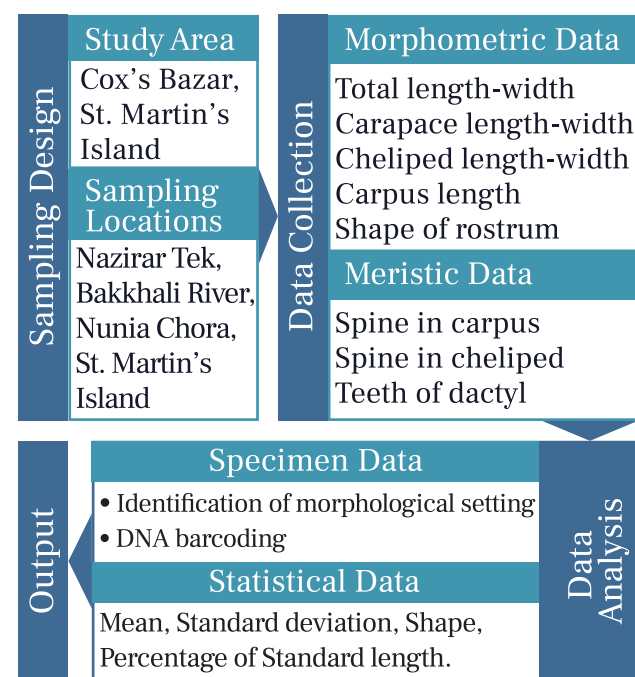


Figure 2 External morphology of Hermit crab

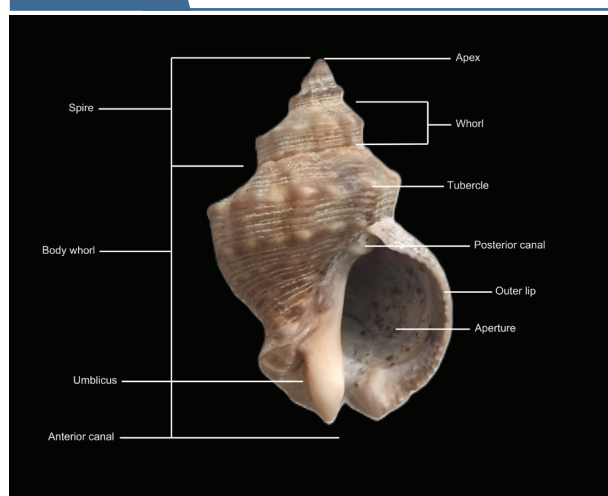


Figure 3 External morphology of gastropod shell

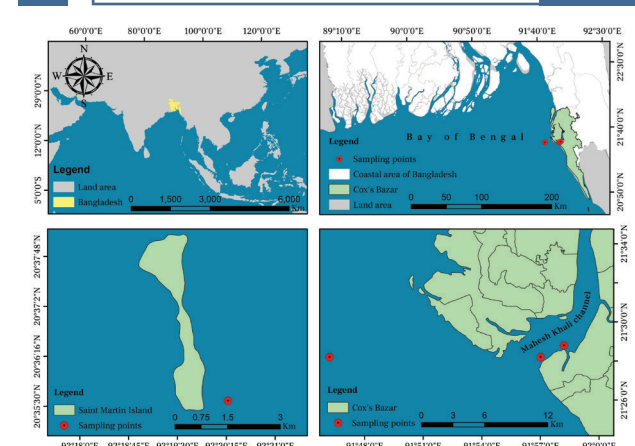
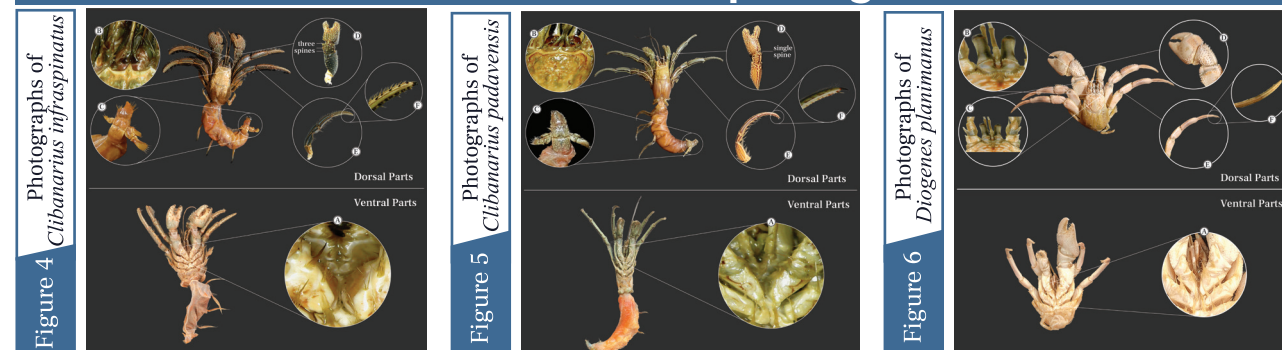


Figure 1 Sampling locations

FINDINGS OF THE ONGOING STUDY

Hermit crabs based on their morphological characteristics



FINDINGS... (CONTD)

The relationship between Hermit crab and Gastropod shells

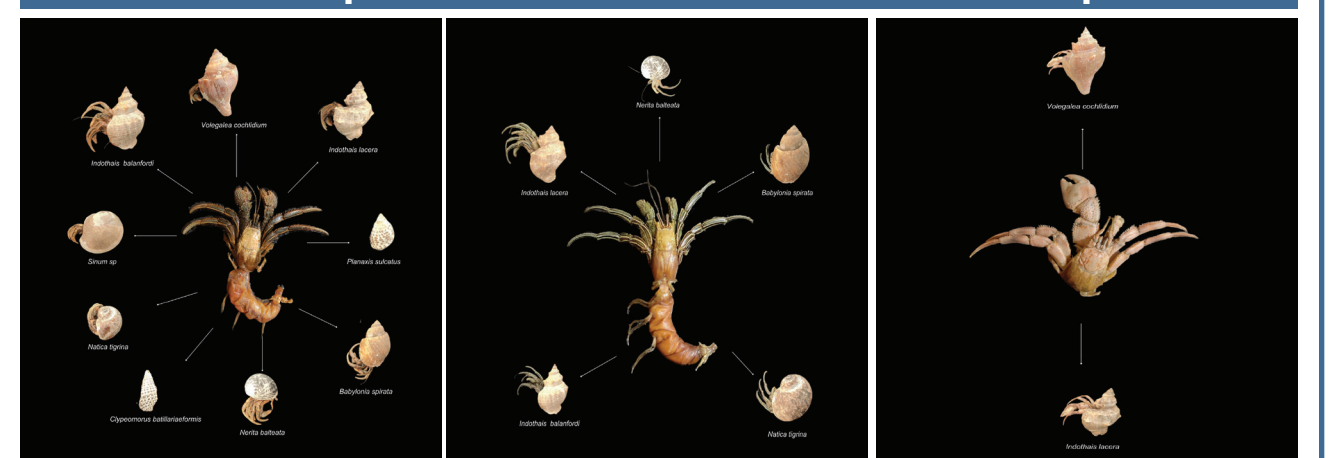


Figure 7 Photographs of *Cibbanarius infraspinus* and gastropod shells
Figure 8 Photographs of *Cibbanarius padavensis* and gastropod shells
Figure 9 Photographs of *Diogenes planimanus* and gastropod shells

Identification of the Gastropod shells

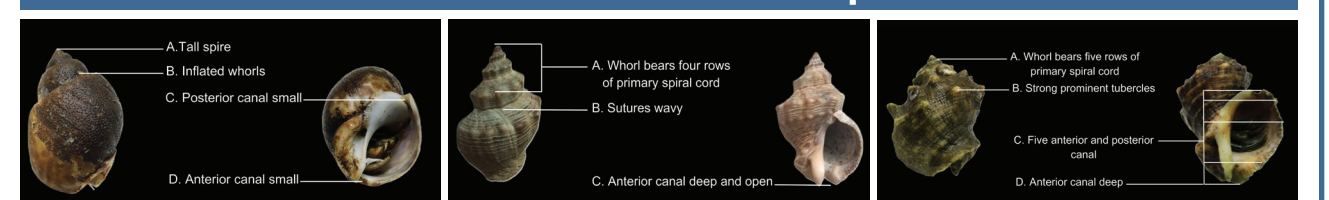


Figure 10 Morphometric features of *Babylonia spirata*
Figure 11 Morphometric features of *Indothais blanfordi*
Figure 12 Morphometric features of *Indothais lacera*

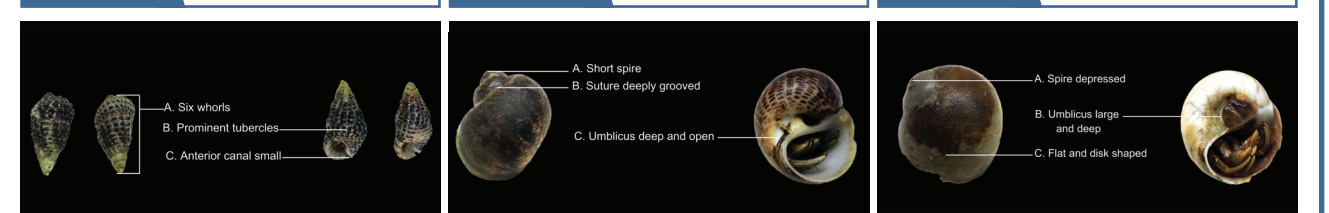


Figure 13 Morphometric features of *Clypeomorus batillariaeformis*
Figure 14 Morphometric features of *Natica tigrina*
Figure 15 Morphometric features of *Sinum* sp.

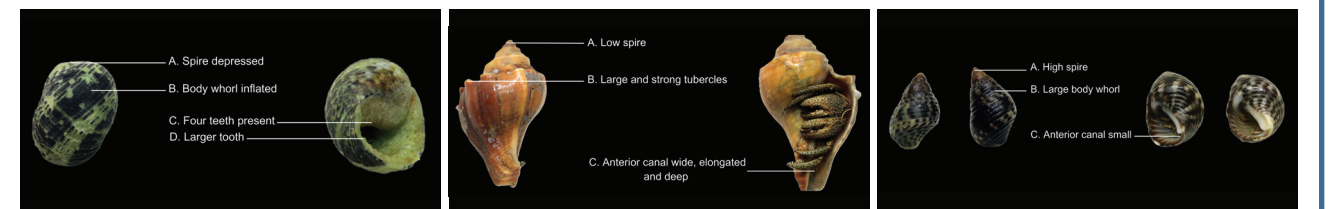


Figure 16 Morphometric features of *Nerita balteata*
Figure 17 Morphometric features of *Volegalea cochlidium*
Figure 18 Morphometric features of *Planaxis sulcatus*

An updated checklist of Hermit crabs from Bangladesh

Genus / species	Habitat	ICUN status
<i>Clibanarius</i> (Dana, 1852)		
1. <i>Clibanarius infraspinus</i> * (Hilgendorf, 1869)	Marine	NE
2. <i>Clibanarius longitarsus</i> (De Haan, 1849)	Brackish & Marine	LC
3. <i>Clibanarius padavensis</i> * (Dana 1852)	Brackish & Marine	NE
<i>Coenobita</i>		
4. <i>Coenobita variabilis</i>	Terrestrial	DD
5. <i>Coenobita violascens</i>	Terrestrial	DD
<i>Diogenes</i> (Dana, 1852)		
6. <i>Diogenes planimanus</i> * (Henderson, 1893)	Marine	NE
<i>Pagurus</i> (Dana, 1852)		
7. <i>Pagurus bernhardus</i>	Brackish & Marine	DD
<i>Parapagurus</i>		
8. <i>Parapagurus nudus</i>	Marine	DD

DD = Data Deficient; LC = Least Concern; NE = Not Evaluated

CONCLUSION

Three new species of hermit crabs have been reported as a new record throughout the coast of Bangladesh. There is also a noted checklist provided for the hermit crab species.

REFERENCES

